

ANALYSIS OF IoT BOTNET STRUCTURE AND BEHAVIOR: A STUDY BASED ON THE HITORI (SLUMX) VARIANT

Ariel Matias Melo Pazmiño

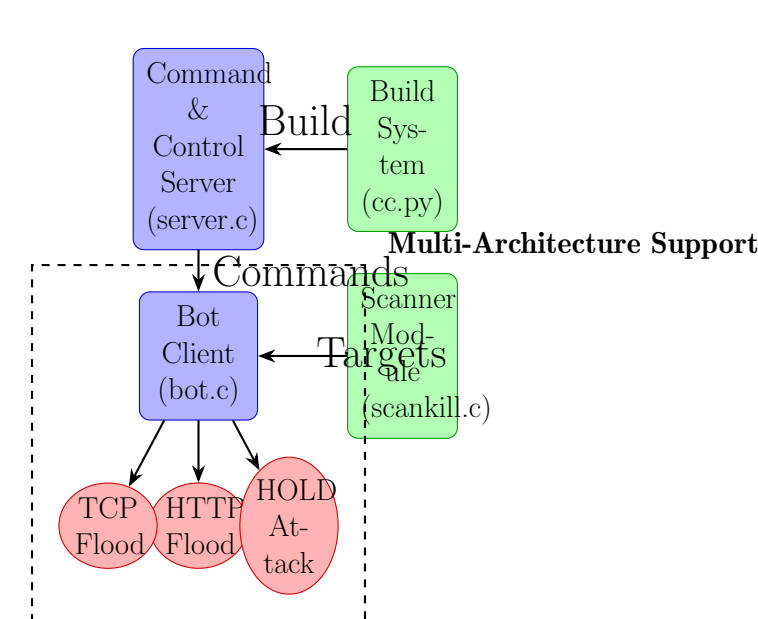
Universidad Internacional del Ecuador - Facultad de Ciencias Técnicas

Abstract

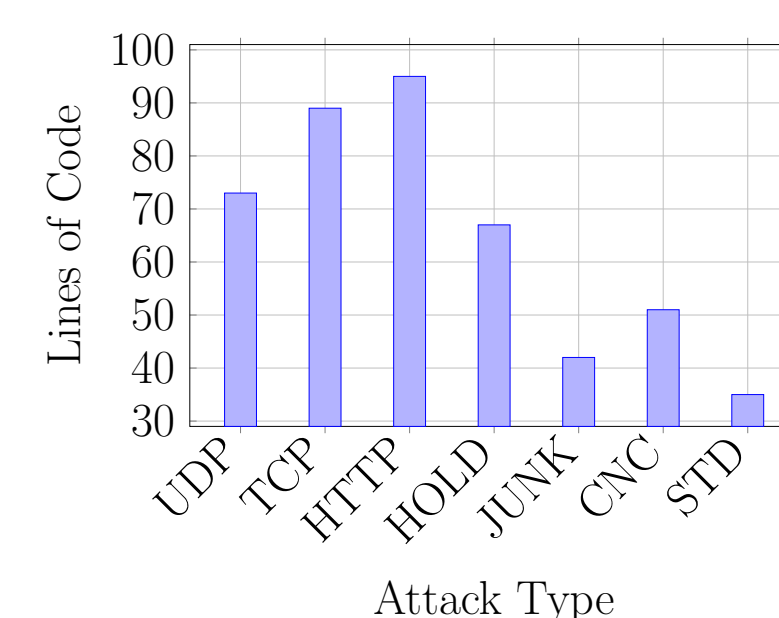
The Internet of Things has fundamentally transformed cybersecurity, introducing billions of connected devices with minimal security measures. This research presents a comprehensive technical analysis of the SlumX botnet (Hitori variant) through static code analysis of the open-source repository.

Key Focus: Understanding the evolution from simple credential attacks to sophisticated multi-architecture IoT malware with advanced evasion capabilities.

System Architecture



Attack Vector Complexity



HTTP-based attacks show highest complexity (95 LOC) due to request crafting and user-agent randomization requirements.

Research Objectives

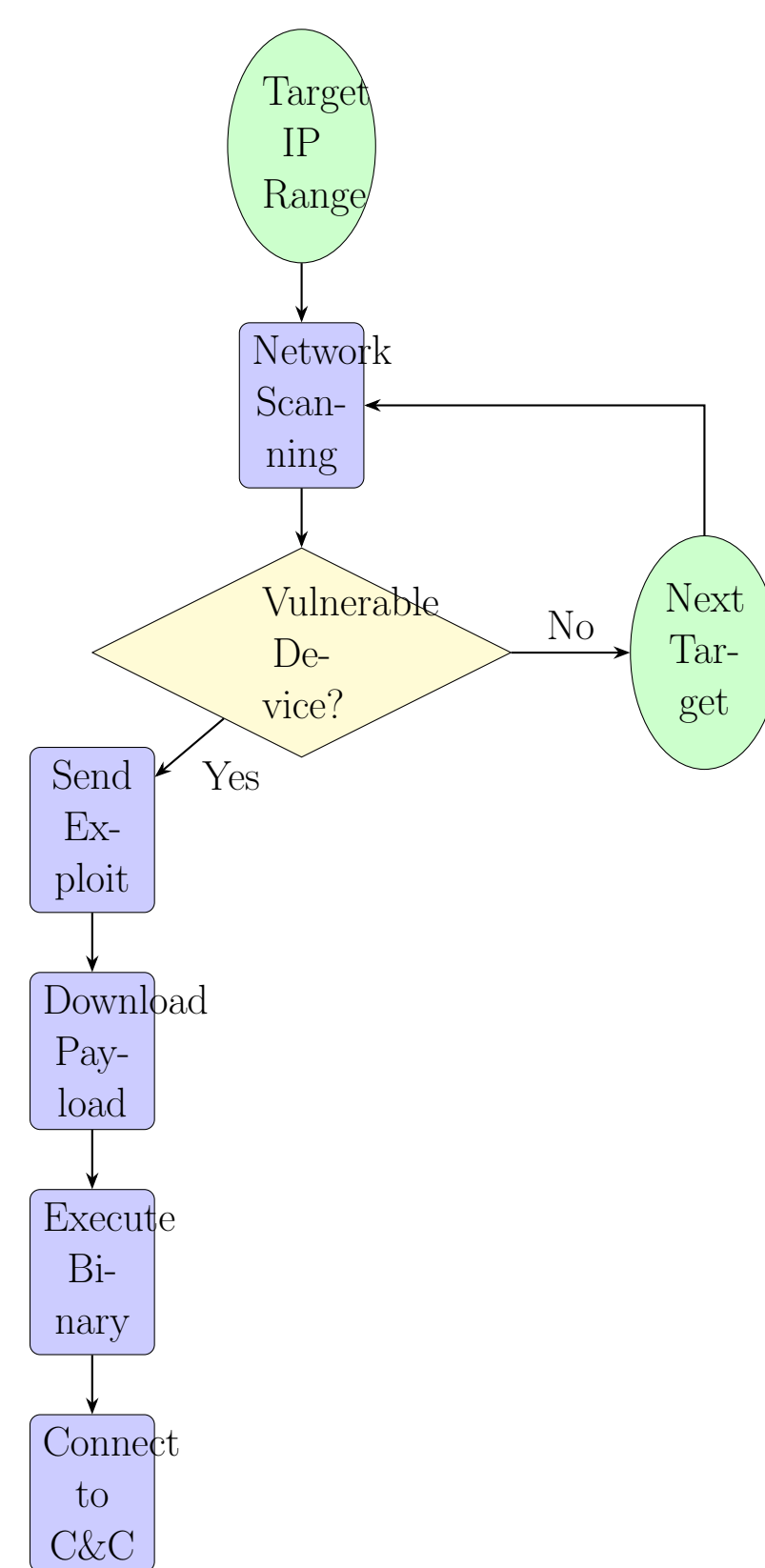
General Objective:

- Conduct in-depth technical analysis of Hitori (SlumX) variant to understand modern IoT botnet threats

Specific Objectives:

- Analyze core architecture and operational framework
- Examine Command & Control infrastructure design
- Identify and categorize DDoS attack vectors
- Evaluate multi-architecture support mechanisms
- Document evasion and anti-analysis techniques
- Assess IoT device scanning and exploitation capabilities

Infection Process Flow



Vulnerability Assessment

Hardcoded C&C Address (Single Point Failure)	Plain Text Communication (Traffic Analysis)
Fixed Port Usage (Network Filtering)	Predictable Binary Names (Signature Detection)

■ High Risk
 ■ Medium Risk
 ■ Low Risk

Key Findings

- Multi-architecture support:** 12 architectures (MIPS, ARM, x86, embedded systems) dramatically expanding attack surface
- Advanced evasion:** Process masquerading, anti-analysis features, IoT-specific hiding techniques
- Targeted exploitation:** Vendor-specific exploits and protocol-based attacks showing evolution from opportunistic to targeted attacks
- Implementation flaws:** Buffer overflow vulnerabilities, hardcoded credentials, predictable attack patterns

Methodology

Static Code Analysis Framework

Analysis of the `mat1520/slumpx` repository using:

- Code Review:** Systematic examination of source files (`bot.c`, `server.c`, `scankill.c`)
- Control Flow Analysis:** Mapping execution paths and decision points
- Threat Modeling:** Assessment of attack capabilities and defensive gaps
- Architecture Analysis:** Cross-platform compilation and targeting strategies

Limitations:

- Static analysis cannot capture all runtime behaviors
- No testing on actual IoT devices
- Temporal snapshot of evolving malware

Multi-Architecture Support

Architecture Target Devices

MIPS	Routers, Network Equipment
ARM	IoT Devices, Cameras
x86	PC-based IoT Systems
PowerPC	Legacy Embedded Systems
SPARC	Enterprise IoT
M68k	Industrial Controllers
SH4	Multimedia Devices

12 supported architectures expanding attack surface across diverse IoT ecosystem.

Conclusions

This analysis reveals significant advancement in IoT malware sophistication, particularly in multi-architecture targeting and evasion capabilities. The findings demonstrate evolution from opportunistic to targeted IoT attacks.

Future Work:

- Dynamic analysis frameworks for multi-architecture IoT malware
- Specialized detection technologies for resource-constrained devices
- Comparative analysis of botnet evolution patterns

Research Paper QR Code

